

From the continuum to discrete substrates: 't Hooft and other turns

Conversation record, 2026-09-28 (user and Claude, with a from-memory answer by Claude Fable). All references below are quoted from memory and have not been verified; they are leads for the prior-art section of the fifth-postulate paper, and no result here is claimed.

Conversation with the user, 2026-09-28 night (not yet in any repo note; references from memory, unverified).

't Hooft (Fable's memory, high/medium confidence): renormalizability and $\beta < 0$ (Marseille 1972); only first two beta coefficients scheme-independent (limits "2 b0 per log" beyond one loop); renormalons (Erice late 1970s) = the atlas exponential ladder, threshold $2 b0 c = 4$ is the gluon-condensate renormalon; instantons $e^{-8 \pi^2/g^2}$, U(1) problem; 't Hooft loop $B(C)$ (NPB 138, 1978) = our mid-edge centre flip; twisted b.c. and e/m flux sectors (NPB 153, 1979), fractional charge $1/N \bmod 1$ ($1/3$ for SU(3), fits the trisection result); the round-12 non-based-cycle Aharonov-Bohm phase is the small-angle part of his flux-sector data (round-14 prompt now asks Astra to state the covariant reference sector by sector); abelian projection 1981; large N 1974. Cellular-automaton interpretation (1988 on, book 2016): $\hbar$ enters as a conversion constant tied to the time grain δt ; the floor comes from information loss (equivalence classes give H bounded below). This is the exact opposite of the user's thesis on the same RG ground: he supplies $\hbar$ from below; we ask whether consistent refinement forces it.

Candidate next Newton premise (proposed, not yet sent to Astra): take 't Hooft's information-loss/equivalence-class construction as the physical premise and ask whether it bounds the recorded Galileo comparison. Contact point: our thermodynamic-records note gave only $\eta \geq A_0 e^{-W/kBT}$, no floor. Put this to an Astra decision round after round 14.

User's remarks: "as soon as you really understand the RG then you don't understand any" (amused); read Wolfram in the eighties thinking "how will he reach a continuum limit", yet the Newton thesis says the continuum limit is problematic from the start, so "a bit bipolar". Claude's reply: same fault line from two sides; the atlas takes the limit and asks what survives.

Wolfram (Caltech ~1979, QCD phenomenology, Fox-Wolfram moments; cellular automata 1981-82; NKS 2002; hypergraph physics 2020). **Cvitanovic** (Kinoshita's group, QED g-2; birdtracks from colour factors, book 2008, useful notation for our SU(N) Brauer-Klimyk sums; chaos via the Feigenbaum-Cvitanovic RG fixed point, ChaosBook).

Other "Damascus" actors listed: divergence-driven: Heisenberg universal length (1938), Snyder quantized spacetime (1947), Yukawa elementary domains, Dirac's lifelong rejection of renormalization. RG masters going deterministic/discrete: T. D. Lee discrete time (1983), Adler trace dynamics (2004), H. B. Nielsen random dynamics, Feynman checkerboard and "Simulating physics with computers" (1982). Geometry to combinatorics: Regge, Penrose spin networks, Finkelstein space-time code (1969), Sorkin causal sets, Ambjorn-Loll, Wheeler "it from bit", von Weizsacker ur-alternatives. **Closest to the Newton thesis:** Gisin and Del Santo (~2019-21), finite information makes classical mechanics indeterministic (denial of the *velocitas ultima* from quantum information) but give no $\hbar$; the fifth-postulate paper should cite and answer them. Proposed paper sentence: field theorists went discrete to escape infinities; RG masters to explain the quantum; Gisin/Del Santo to break determinism; the user's thesis keeps the continuum and shows consistent refinement demands the constant Newton's limit dropped.

Related: STATE, fifth postulate, stochastic route, thermodynamic records.

't Hooft's automaton against Zeno, the atomists and Newton

Added 2026-09-29 (Claude), at the user's request. The automaton papers are identified by DOI (Crossref-verified, metadata only): the 1988 equivalence relations, the 1999 dissipative deterministic system and the 2016 book. What follows is a reading of his position as Fable and Claude remember it, set against the classics entries of the Planck paper §§7 and 9.

His commitments. Reality is a deterministic automaton with a universal time step δt . A reversible automaton is a permutation of states, so its evolution operator has eigenvalues on the unit circle and energies defined only modulo $2\pi\hbar/\delta t$: $\hbar$ is the conversion between the step and an energy, and the spectrum has no ground state. Information loss, many states merging into one equivalence class, is what he invokes to obtain a Hamiltonian bounded below. Quantum mechanics is the description of the classes; Bell correlations are paid for by superdeterminism.

- **Zeno and Diodorus.** At a tick the automaton holds a configuration and motion exists only as the update between ticks: Diodorus’s “a thing never is moving, but it has moved”, made into dynamics. The Planck paper keeps Zeno’s conclusion for local records only, below the window $\tau_{\text{arrow}} = 8z^2\kappa/(mv^2)$, which depends on the body and the confidence. The automaton makes the arrow exact at every scale below δt and for every body.
- **Democritus and the kalam atom.** The automaton’s cells and ticks are atoms of place and time, the kalam position on time and a stepped cone in Democritus’s dilemma. The Planck paper denies the universal time atom on the evidence of records: its mesh moves with F , m and the confidence, a dynamical resolution. The two are compatible: a universal grain far below every mark mesh leaves Theorem 9 untouched. They differ in status: the grain is ontological, the mesh is epistemic.
- **Epicurus.** Continuity presented to sense, succession below it, and a warning against extrapolating continuity downward: this is ’t Hooft’s picture with the Planck time in place of the threshold of sense. Epicurus is the closest ancient statement of it, as he is of ours; the difference is where the boundary is drawn and whether it is derived.
- **Newton.** The automaton removes the *velocitas ultima* by removing the limit: velocities are finite differences and no instant carries one. That is a third denial of joint determinacy, beside quantum noncommutativity (Theorem A) and stochastic paths (the stochastic route), and it keeps determinism. Its $\hbar$ is a unit conversion, so the automaton alone supplies a grain and no action floor on records: a reversible automaton is a discrete Liouville dynamics, and Theorem I of the unit-and-indeterminacy note gives classical readouts no floor. The floor, in his construction, rides on information loss, and the thermodynamic-records note finds that loss yields a trade-off $\eta \geq A_0 e^{-W/k_B T}$ and no floor.

The sharp question this leaves, the candidate Newton premise above: does coarse-graining by equivalence classes, of the kind that gives his Hamiltonian a ground state, force a positive floor on the recorded inertial–parabola comparison, or only a trade-off? The repository’s current results point to the second; a theorem either way would place the automaton exactly relative to the thesis that quantization is a consistency condition of the continuum limit.

Later the same night: superdeterminism and Dedekind’s cut

Superdeterminism (user: “classical mechanics is accidentally superdeterminist too, and we think it has some mistake somewhere”). From memory: Bell called the independence of settings and hidden variables “free variables” (exchange with Shimony, Horne and Clauser, around 1976–77) and named the escape “super-deterministic” in a 1985 BBC interview (*The Ghost in the Atom*, 1986); Brans gave a model in 1988; ’t Hooft adopted it in the 2000s; Hossenfelder and Palmer revived it in 2020. The link the user and Claude found interesting is developed in the superdeterminism note: for chaotic setting devices, a superdeterministic correlation must be stored in structure of the initial data finer than any fixed resolution, so a floor gives it an Ehrenfest-type lifetime and $\hbar \rightarrow 0$ reopens the loophole.

Dedekind’s cut (user: “Dedekind cuts are peculiar, as they define a point as a segment, really”). Dedekind (1872) defines a real number by the two segments of rationals it separates: the point is given by what lies on either side, and fixing it needs infinitely many comparisons. The intuitionists, and after them Gisin and Del Santo (from memory: finite-information quantities

whose digits are fixed progressively, around 2019–21), refuse the completed cut: a real is only ever a nested sequence of segments, never a point. The repository has a quantitative form of the remark. Locating an instant by nested cuts spends Newton’s cell action additively (cut-measure note, Theorem 2), and with a floor κ per exhibited cut only K_τ/κ cuts can be exhibited (Corollary 5): a recorded instant is a segment of length of order the mark mesh τ_* . Dedekind’s point is the completed limit of that sequence, which is Newton’s side in the scholium closing Book I (Euclid X against least magnitudes); the recorded point is the unfinished one.

Connes (user: “Connes more or less exits the dilemma by having $dx = [D, X]$ ”). From memory: in a spectral triple the differential of f is $[D, f]$, infinitesimals are compact operators, the line element is $ds = D^{-1}$, and distance needs no paths, $d(p, q) = \sup\{|f(p) - f(q)| : \|[D, f]\| \leq 1\}$. The completed point is never formed, and the velocity is an operator relation, as in Heisenberg’s $\dot{x} = (i/\hbar)[H, x]$. Two refinements: in the commutative case $[D, f]$ commutes with every function, so joint determinacy holds and Newton is recovered; noncommutativity remains the premise, the fork of Theorem A of the fifth-postulate note. With $p = \hbar D$ the line element D^{-1} is $\hbar/p$, the de Broglie wavelength operator, so $\hbar$ is again a conversion constant. The tangent-groupoid note already holds the gluing of pairs at $\hbar > 0$ (the segment) to tangent vectors at $\hbar = 0$ (the point). A candidate Newton test: whether the repeated cutting of a Galileo cell extends continuously to the $\hbar = 0$ fibre.

The test, answered (Claude, 2026-09-29). It reduces to an order of limits. Under the path measure of the cut-measure note, a Galileo cell of duration τ distinguishes constant force from inertia with error at most ϵ iff $K_\tau = F^2\tau^3/(24m) \geq 2z_{1-\epsilon}^2\hbar$ (Proposition 6). Recorded chords therefore have $\tau \geq \tau_* = (48z_{1-\epsilon}^2 m\hbar/F^2)^{1/3}$, the mark mesh, and in the groupoid topology a sequence (x, y, τ) of such chords reaches the fibre $TM \times \{0\}$ only if $\hbar/K_\tau \rightarrow 0$ along it. At fixed $\hbar > 0$ the recorded arrows keep the distance τ_* from Newton’s fibre; on the branch $\hbar = 0$, and on every path with $\hbar = o(F^2\tau^3/m)$, they reach it continuously. Repeated cutting is consistent on both sides, so the test restates the fork of Theorem A and supplies no necessity. The decision round of 2026-09-29 (Astra) reached the same verdict without the computation and chose cell 2.

Consequence for STATE

None. The Astra decision round of 2026-09-29 weighed the information-loss premise of ’t Hooft’s equivalence classes, the tangent-groupoid test and cell 4, and chose cell 2: the equivalence classes still lack a stated link to a lower bound on recorded Galileo areas, and the groupoid test, answered above, restates the fork.